JOINT ADMISSION TEST FOR MASTERS • IIT JAM MATHEMATICS (MA)

Integral Calculus (2015–2021)

Category: Multivariable Calculus and Differential Equations • Official Mock Test Series

TOTAL QUESTIONS	TOTAL MARKS	TEST DURATION	PAPER PATTERN
31	51 M	93 Min	MCQ • MSQ • NAT (2015–20

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	15	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	2	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	14	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

1. **Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
2. **Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
3. **Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
4. **Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
5. **Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
6. **Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at **vaibhavgeometer.github.io**.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2021 • Q43 • ID: JAM_2021_Q43) NAT +1 M Neg: Nil (0)

Q. 43 Let $B = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 \leq 1\}$ and define $u(x, y, z) = \sin\left((1 - x^2 - y^2 - z^2)^2\right)$ for $(x, y, z) \in B$. Then the value of

$$\iiint_B \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}\right) dxdydz$$

is _____.

Question 2 (JAM 2021 • Q5 • ID: JAM_2021_Q5) MCQ +1 M Neg: −0.33

Q. 5 Let p and t be positive real numbers. Let D_t be the closed disc of radius t centered at $(0, 0)$, i.e., $D_t = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq t^2\}$. Define

$$I(p, t) = \iint_{D_t} \frac{dxdy}{(p^2 + x^2 + y^2)^p}.$$

Then $\lim_{t\to\infty} I(p, t)$ is finite

(A) only if $p > 1$.

(B) only if $p = 1$.

(C) only if $p < 1$.

(D) for no value of p.

Question 3 (JAM 2020 • Q47 • ID: JAM_2020_Q47) NAT +1 M Neg: Nil (0)

Q. 47 If $\int\limits_0^1 \int\limits_{2y}^2 e^{x^2} dxdy = k(e^4 - 1)$, then k equals _______.

Question 4 (JAM 2020 • Q38 • ID: JAM_2020_Q38) MSQ +2 M Neg: Nil (0)

Q. 38 Let S be that part of the surface of the paraboloid $z = 16 - x^2 - y^2$ which is above the plane $z = 0$ and D be its projection on the xy-plane. Then the area of S equals

(A) $\iint\limits_D \sqrt{1 + 4(x^2 + y^2)}\, dxdy$

(B) $\iint\limits_D \sqrt{1 + 2(x^2 + y^2)}\, dxdy$

(C) $\int\limits_0^{2\pi} \int\limits_0^4 \sqrt{1 + 4r^2}\, drd\theta$

(D) $\int\limits_0^{2\pi} \int\limits_0^4 \sqrt{1 + 4r^2}\, rdrd\theta$

Question 5 (JAM 2020 • Q25 • ID: JAM_2020_Q25) **MCQ** **+2 M** **Neg: −0.67**

Q. 25 Let S be the part of the cone $z^2 = x^2 + y^2$ between the planes $z = 0$ and $z = 1$. Then the value of the surface integral $\iint\limits_S (x^2 + y^2)\, dS$ is

(A) π (B) $\frac{\pi}{\sqrt{2}}$ (C) $\frac{\pi}{\sqrt{3}}$ (D) $\frac{\pi}{2}$

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Question 6 (JAM 2020 • Q24 • ID: JAM_2020_Q24) **MCQ** **+2 M** **Neg: −0.67**

Q. 24 The value of the triple integral $\iiint\limits_V (x^2y + 1)\, dxdydz$, where V is the region given by $x^2 + y^2 \leq 1, 0 \leq z \leq 2$ is

(A) π (B) 2π (C) 3π (D) 4π

Question 7 (JAM 2020 • Q14 • ID: JAM_2020_Q14) **MCQ** **+2 M** **Neg: −0.67**

Q. 14 Let $a \in \mathbb{R}$. If $f(x) = \begin{cases} (x+a)^2, & x \leq 0 \\ (x+a)^3, & x > 0, \end{cases}$

then

(A) $\frac{d^2 f}{dx^2}$ does not exist at $x = 0$ for any value of a

(B) $\frac{d^2 f}{dx^2}$ exists at $x = 0$ for exactly one value of a

(C) $\frac{d^2 f}{dx^2}$ exists at $x = 0$ for exactly two values of a

(D) $\frac{d^2 f}{dx^2}$ exists at $x = 0$ for infinitely many values of a

$$\Big(x^2 \sin \tfrac{1}{x} + y^2 \sin \tfrac{1}{y}, \qquad xy \neq 0$$

Question 8 (JAM 2019 • Q51 • ID: JAM_2019_Q51) NAT +2 M Neg: Nil (0)

Q. 51 – Q. 60 carry two marks each.

Q.51 The volume of the solid of revolution of the loop of the curve $y^2 = x^4(x+2)$ about the x-axis (round off to 2 decimal places) is ___________

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Question 9 (JAM 2019 • Q50 • ID: JAM_2019_Q50) NAT +1 M Neg: Nil (0)

Q.50 The volume of the solid bounded by the surfaces $x = 1 - y^2$ and $x = y^2 - 1$, and the planes $z = 0$ and $z = 2$ (round off to 2 decimal places) is _____________

Q. 51 – Q. 60 carry two marks each

Question 10 (JAM 2019 • Q49 • ID: JAM_2019_Q49) NAT +1 M Neg: Nil (0)

Q.49 The value of the integral

$$\int_{-1}^{1}\int_{-1}^{1} |x+y| \, dx \, dy$$

(round off to 2 decimal places) is ___________

Question 11 (JAM 2019 • Q26 • ID: JAM_2019_Q26) MCQ +2 M Neg: −0.67

Q.26 The tangent line to the curve of intersection of the surface $x^2 + y^2 - z = 0$ and the plane $x + z = 3$ at the point $(1, 1, 2)$ passes through

(A) $(-1, -2, 4)$ (B) $(-1, 4, 4)$ (C) $(3, 4, 4)$ (D) $(-1, 4, 0)$

Question 12 (JAM 2019 • Q10 • ID: JAM_2019_Q10) MCQ +1 M Neg: −0.33

Q.10 The area of the surface generated by rotating the curve $x = y^3,\ 0 \le y \le 1$, about the y-axis, is

(A) $\frac{\pi}{27} 10^{3/2}$ (B) $\frac{4\pi}{3}(10^{3/2} - 1)$ (C) $\frac{\pi}{27}(10^{3/2} - 1)$ (D) $\frac{4\pi}{3} 10^{3/2}$

Q. 11 – Q. 30 carry two marks each.

Question 13 (JAM 2018 • Q58 • ID: JAM_2018_Q58) NAT +2 M Neg: Nil (0)

Q.58 The area of the parametrized surface

$S = \{((2 + \cos u)\cos v,\ (2 + \cos u)\sin v,\ \sin u) \in \mathbb{R}^3 \mid 0 \le u \le \frac{\pi}{2},\ 0 \le v \le \frac{\pi}{2}\}$

is ____________ (correct up to two decimal places).

Question 14 (JAM 2018 • Q56 • ID: JAM_2018_Q56) NAT +2 M Neg: Nil (0)

Q.56 The value of the integral

$$\int_0^1 \int_x^1 y^4 e^{xy^2}\, dy\, dx$$

is ______________ (correct up to three decimal places).

Question 15 (JAM 2018 • Q54 • ID: JAM_2018_Q54) NAT +2 M Neg: Nil (0)

Q.54 If the volume of the solid in $\mathbb{R}^3$ bounded by the surfaces

$$x = -1, \quad x = 1, \quad y = -1, \quad y = 1, \quad z = 2, \quad y^2 + z^2 = 2$$

is $\alpha - \pi$, then $\alpha =$ ________________

Question 16 (JAM 2018 • Q20 • ID: JAM_2018_Q20) MCQ +2 M Neg: −0.67

Q.20 If $\vec{F}(x, y) = (3x - 8y)\hat{\imath} + (4y - 6xy)\hat{\jmath}$ for $(x, y) \in \mathbb{R}^2$, then $\oint_C \vec{F} \cdot d\,\vec{r}$, where C is the boundary of the triangular region bounded by the lines $x = 0, y = 0$ and $x + y = 1$ oriented in the anti-clockwise direction, is

(A) $\frac{5}{2}$ (B) 3 (C) 4 (D) 5

Question 17 (JAM 2018 • Q10 • ID: JAM_2018_Q10) MCQ +1 M Neg: −0.33

Q.10 Let $s_n = 1 + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!}$ for $n \in \mathbb{N}$. Then which one of the following is TRUE for the sequence $\{s_n\}_{n=1}^{\infty}$

(A) $\{s_n\}_{n=1}^{\infty}$ converges in $\mathbb{Q}$

(B) $\{s_n\}_{n=1}^{\infty}$ is a Cauchy sequence but does not converge in $\mathbb{Q}$

(C) the subsequence $\{s_{k^n}\}_{n=1}^{\infty}$ is convergent in $\mathbb{R}$, only when k is even natural number

(D) $\{s_n\}_{n=1}^{\infty}$ is not a Cauchy sequence

Question 18 (JAM 2017 • Q56 • ID: JAM_2017_Q56) NAT +2 M Neg: Nil (0)

Q.56 For a real number x, define $\lceil x \rceil$ to be the smallest integer greater than or equal to x. Then

$$\int_0^1 \int_0^1 \int_0^1 \left(\lceil x \rceil + \lceil y \rceil + \lceil z \rceil \right) \, dx\, dy\, dz = \_\_\_\_\_\_$$

Question 19 (JAM 2017 • Q32 • ID: JAM_2017_Q32) MSQ +2 M Neg: Nil (0)

Q.32 The volume of the solid

$$\left\{(x,y,z)\in\mathbb{R}^3 \,\middle|\, 1\le x\le 2,\quad 0\le y\le \frac{2}{x},\quad 0\le z\le x\right\}$$

is expressible as

(A) $\int_1^2\int_0^{2/x}\int_0^x dz\,dy\,dx$

(B) $\int_1^2\int_0^x\int_0^{2/x} dy\,dz\,dx$

(C) $\int_0^2\int_1^z\int_0^{2/x} dy\,dx\,dz$

(D) $\int_0^2\int_{\max\{z,1\}}^2\int_0^{2/x} dy\,dx\,dz$

Question 20 (JAM 2017 • Q16 • ID: JAM_2017_Q16) MCQ +2 M Neg: −0.67

Q.16 The area of the surface $z=\frac{xy}{3}$ intercepted by the cylinder $x^2+\ y^2\le 16$ lies in the interval

(A) $(20\pi, 22\pi]$ (B) $(22\pi, 24\pi]$ (C) $(24\pi, 26\pi]$ (D) $(26\pi, 28\pi]$

Question 21 (JAM 2017 • Q6 • ID: JAM_2017_Q6) MCQ +1 M Neg: −0.33

Q.6

$$\int_0^1\int_x^1 \sin(y^2)dy\,dx\ =$$

(A) $\frac{1+\cos\ 1}{2}$ (B) $1-\cos 1$ (C) $1+\cos 1$ (D) $\frac{1-\cos\ 1}{2}$

Question 22 (JAM 2016 • Q57 • ID: JAM_2016_Q57) NAT +2 M Neg: Nil (0)

Q.57 The value of the double integral

$$\int_0^{\pi}\int_0^{x}\frac{\sin y}{\pi-y}\,dy\,dx$$

is ____________

Question 23 (JAM 2016 • Q52 • ID: JAM_2016_Q52) NAT +2 M Neg: Nil (0)

Q.52 Let R be the region enclosed by $x^2 + 4y^2 \geq 1$ and $x^2 + y^2 \leq 1$. Then the value of

$$\iint_R |xy|\, dx\, dy \quad \text{is } \_\_\_\_\_\_\_\_$$

Question 24 (JAM 2016 • Q30 • ID: JAM_2016_Q30) MCQ +2 M Neg: −0.67

Q.30 Let $f: \mathbb{R} \to \mathbb{R}$ be defined as

$$f(x) = \begin{cases} x(1 + x^{\alpha} \sin(\ln x^2)) & \text{if} \quad x \neq 0 \\ 0 & \text{if} \quad x = 0 \end{cases}.$$

Then, at $x = 0$, the function f is

(A) continuous and differentiable when $\alpha = 0$
(B) continuous and differentiable when $\alpha > 0$
(C) continuous and differentiable when $-1 < \alpha < 0$
(D) continuous and differentiable when $\alpha < -1$

SECTION - B

MULTIPLE SELECT QUESTIONS (MSQ)

Q. 31 – Q. 40 carry two marks each.

Question 25 (JAM 2016 • Q20 • ID: JAM_2016_Q20) MCQ +2 M Neg: −0.67

Q.20 Let S be the surface of the cone $z = \sqrt{x^2 + y^2}$ bounded by the planes $z = 0$ and $z = 3$. Further, let C be the closed curve forming the boundary of the surface S. A vector field $\vec{F}$ is such that $\nabla \times \vec{F} = -x\hat{\imath} - y\hat{\jmath}$. The absolute value of the line integral $\oint_c \vec{F} \cdot d\vec{r}$, where $\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$ and $r = |\vec{r}|$, is

(A) 0 (B) 9π (C) 15π (D) 18π

Question 26 (JAM 2016 • Q10 • ID: JAM_2016_Q10) MCQ +1 M Neg: −0.33

Q.10 Let S be a closed subset of $\mathbb{R}$, T a compact subset of $\mathbb{R}$ such that $S \cap T \neq \phi$. Then $S \cap T$ is

(A) closed but not compact
(B) not closed
(C) compact
(D) neither closed nor compact

Q. 11 – Q. 30 carry two marks each.

Question 27 (JAM 2015 • Q58 • ID: JAM_2015_Q58) NAT +2 M Neg: Nil (0)

Q.18

Let ℓ be the length of the portion of the curve $x = x(y)$ between the lines $y = 1$ and $y = 3$, where $x(y)$ satisfies

$$\frac{dx}{dy} = \frac{\sqrt{1+y^2+y^4}}{y}, \qquad x(1) = 0$$

The value of ℓ, correct upto three decimal places, is ______________

Question 28 (JAM 2015 • Q55 • ID: JAM_2015_Q55) NAT +2 M Neg: Nil (0)

Q.15

Let $f\colon \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$f(x,y) = \begin{cases} \dfrac{y}{\sin y}, & y \neq 0 \\ 1, & y = 0 \end{cases}$$

Then the integral

$$\frac{1}{\pi^2}\int_{x=0}^{1}\int_{y=\sin^{-1} x}^{\frac{\pi}{2}} f(x,y)\, dy\, dx$$

correct upto three decimal places, is ______________

Question 29 (JAM 2015 • Q42 • ID: JAM_2015_Q42) NAT +1 M Neg: Nil (0)

Q.2

Let S be the portion of the surface $z = \sqrt{16 - x^2}$ bounded by the planes $x = 0$, $x = 2$, $y = 0$, and $y = 3$. The surface area of S, correct upto three decimal places, is ______________

Question 30 (JAM 2015 • Q26 • ID: JAM_2015_Q26) MCQ +2 M Neg: −0.67

Q.26 The area of the planar region bounded by the curves $x = 6y^2 - 2$ and $x = 2y^2$ is

(A) $\frac{\sqrt{2}}{3}$ (B) $\frac{2\sqrt{2}}{3}$ (C) $\frac{4\sqrt{2}}{3}$ (D) $\sqrt{2}$

Question 31 (JAM 2015 • Q10 • ID: JAM_2015_Q10) MCQ +1 M Neg: −0.33

Q.10 Let A be a nonempty subset of $\mathbb{R}$. Let $I(A)$ denote the set of interior points of A. Then $I(A)$ can be

(A) empty
(B) singleton
(C) a finite set containing more than one element
(D) countable but not finite

Q. 11 – Q. 30 carry two marks each.

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Integral Calculus (2015–2021) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	NAT	+1	**0**
2	MCQ	+1	**A**
3	NAT	+1	**0.25 to 0.25**
4	MSQ	+2	**A;D**
5	MCQ	+2	**B**
6	MCQ	+2	**B**
7	MCQ	+2	**A**
8	NAT	+2	**6.60 to 6.80**
9	NAT	+1	**5.30 to 5.50**
10	NAT	+1	**2.60 to 2.70**
11	MCQ	+2	**B**
12	MCQ	+1	**C**
13	NAT	+2	**6.30 to 6.70**
14	NAT	+2	**0.230 to 0.250**
15	NAT	+2	**5.99 to 6.01**
16	MCQ	+2	**B**
17	MCQ	+1	**B**
18	NAT	+2	**2.9 to 3.1**
19	MSQ	+2	**A, B, D**
20	MCQ	+2	**A**
21	MCQ	+1	**D**
22	NAT	+2	**2.0 to 2.0**
23	NAT	+2	**0.35 to 0.4**
24	MCQ	+2	**MTA**
25	MCQ	+2	**MTA**
26	MCQ	+1	**C**
27	NAT	+2	**5.09 to 5.10**
28	NAT	+2	**0.125**
29	NAT	+1	**6.28 to 6.29**
30	MCQ	+2	**C**
31	MCQ	+1	**A**

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt (-1/3 for 1-mark questions, −2/3 for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., [a, b]) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.